

Rachakonda Abhinay

AI/ML Engineer

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SKILLS

Programming & Libraries

Python,BeautifulSoup, NumPy, Pandas, Matplotlib, Seaborn, NLTK, Scikit-learn, OpenCV, TensorFlow, PyTorch.

Machine Learning

Supervised & Unsupervised Learning, Feature Engineering, Data Preprocessing, Hyperparameter Tuning, Model Evaluation.

Deep Learning

ANN, CNN, RNN,LSTM,GRU

Generative AI and Agent Frameworks

Transformers architecture ,LangChain, LangGraph, CrewAI, Agno, AutoGen, DeepAgents,Hugging Face Fine-Tuning, Prompt Engineering,Agentic_RAG

Deployment & MLOps:

FastAPI, Docker, MLflow,AWS EC2, Model & Application Deployment

EXPERIENCE

Geeth LLC — AI/ML Engineer 🔗

Feb 2025 – Feb 2026

Remote,Waukesha,
Wisconsin,Usa

- Architected an **Agentic RAG pipeline** for multi-format document QA using FAISS and LLMs, improving retrieval efficiency by ~90%.
- Implemented an **AI-powered ATS Resume Screening System** for resume ingestion and JD-based scoring, with LLM-generated interview questions, reducing manual effort by ~95%.
- Engineered a **Voice Assistant system** integrating speech-to-text → LLM → text-to-speech pipeline for real-time interaction.
- Created a **Video Summarization pipeline** using Whisper transcription and LLM-based summarization for long-form content.
- Integrated **Plaid APIs via FastAPI microservices** to securely retrieve financial data (transactions, balances, identity), deployed using Docker.
- Designed an **AI Decision Intelligence Platform** for automated data profiling, visualization, and insight generation.
- Constructed an **ETL pipeline** for financial data extraction and reconciliation using OCR and rule-based matching.
- Orchestrated a **multi-provider LLM platform** with MongoDB-based conversation logging and prompt optimization.
- Established a **LoRA-based fine-tuning pipeline** for code generation models using Transformers and PyTorch.

Innomatics Research Labs - Data Science Intern 🔗

Jan 2024 – Mar 2024

Hyderabad, India

- Built an AI-powered **Code Reviewer** using Streamlit and OpenAI APIs for automated code analysis.
- Developed a **Conversational Data Science Tutor** using Gemini for interactive ML explanations.
- Implemented **RAG pipelines** using LangChain to improve knowledge retrieval.
- Optimized a **video subtitle search engine** using NLP techniques.
- Set up **ML pipelines** with MLflow and Prefect for experiment tracking and model management.
- Deployed a **sentiment analysis system** using Flask on AWS EC2.

EDUCATION

OSMANIA UNIVERSITY,MBA

MANTRA BUSSINESS SCHOOL OF MANAGEMENT

CGPA: 86%

2023 – 2025

Hyderabad, Telangana

CERTIFICATES

Exploratory Data Analysis using
python Libraries

Advance Machine Learning and
Generative Ai internship
program [↗](#)

Advance Data Analysis internship
program
innomatics research labs

Machine Learning using python
Libraries of IBM [↗](#)

COURSES

Data Science - innomatics research labs [↗](#)

Jun 2023 – Jun 2024
Hyderabad, India

PROJECTS

Data analysis on "trips and travel.com" company

Skills: Pandas, Numpy, matplotlib, seaborn

- Spearheaded the development of a sustainable business model for "Trips & Travel .Com", integrating five existing packages and introducing new wellness options.
- Conducted thorough exploratory data analysis, leading to the creation of a targeted digital marketing strategy. Utilized social media, email marketing, and personalized advertising to engage potential customers and efficiently promote new packages, informed by data-driven insights.

Web Scraping and Exploratory Data Analysis

- Scraped new and used car datasets using BeautifulSoup and structured them using Pandas.
- Performed exploratory data analysis and statistical modeling to uncover pricing trends.
- Created visual dashboards using Matplotlib and Seaborn for insights.
- **Technologies:** Python, BeautifulSoup, Pandas, Matplotlib, Seaborn

Multimodal RAG System

- Built an advanced agentic multimodal system capable of indexing and querying diverse data sources, including structured and unstructured formats.
- Implemented hybrid retrieval by combining semantic search (vector-based) with keyword-based search to significantly improve retrieval accuracy and relevance.
- Utilized MongoDB for efficient storage and management of metadata, document chunks, and indexing references, enabling fast and scalable data access.
- Integrated Groq LLM for high-speed, context-aware response generation, ensuring low latency and high-quality outputs.
- Applied RAG evaluation techniques inspired by RAGAS to measure key performance.

Technologies: Python, LangChain, FAISS, MongoDB, Groq LLM, FastAPI, OCR

Movie Recommendation System using Machine Learning

- Developed a content-based movie recommendation engine using the TMDB dataset.
- Performed NLP preprocessing including tokenization, stopword removal, stemming, and lemmatization using NLTK.
- Extracted textual features using CountVectorizer and computed similarity using Cosine Similarity.
- Built and deployed an interactive Streamlit web application on AWS EC2.
- **Technologies:** Python, Pandas, NLTK, Scikit-learn, CountVectorizer, Cosine Similarity, Streamlit, AWS EC2

Lung Cancer Detection Using VGG16

- Developed a deep learning model to classify lung cancer types including lung squamous cell carcinoma, lung adenocarcinoma, and normal lung tissue using histopathological images.
- Applied image preprocessing and augmentation using ImageDataGenerator. Leveraged a pretrained VGG16 model with frozen base layers and added custom dense layers for classification.
- Trained the model using categorical cross-entropy loss with the Adam optimizer and evaluated performance using confusion matrix and classification reports.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn

PUBLICATIONS

Navigating the GenAI Frontier: Transformers, GPT, and the Path to Accelerated Innovation [↗](#)

Medium

- Explored the evolution of **Generative AI** from traditional **Seq2Seq models** to advanced **Transformers** like **GPT**.
- Discussed key concepts such as **attention mechanisms**, **self-attention**, and **autoregressive language modeling**.

- Highlighted the transformative impact of these technologies on **artificial intelligence**.

OBJECTIVE

Generative AI Engineer with hands-on experience in building RAG pipelines, LLM-based applications, and deploying scalable AI systems using FastAPI and AWS. Passionate about solving real-world problems using AI